

Audio-Visual Speech Enhancement System for Hearing Impaired and Deaf Users in Multi-Speaker Environments

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Abstract - When many people talk at the same time in a noisy place, it becomes very hard to understand what one person is saying. This is often called the "cocktail party problem." Systems that only use audio struggle a lot in such situations. This is especially difficult for deaf or hard-of-hearing people, who depend on watching lips to understand speech. A better approach is to use both audio and video together - this is called Audio-Visual (AV) speech enhancement. This survey looks at several deep learning methods, starting from a well-known base paper by Ephrat et al. (2018) up to newer works. We found that current systems have some important gaps - they cannot handle many speakers at once, they break down when the lip view is blocked, and most are too slow to run on small-scale devices. Our project aims to fix these problems.

Keywords - audio-visual speech enhancement, cocktail party problem, deep learning, hearing impaired, lip reading, multi-speaker separation, real-time processing

I. INTRODUCTION

Speech enhancement means cleaning up speech so it is easier to hear and understand, especially in noisy places where many people are talking. For deaf users, being able to see the speaker's lips is very important. That is why Audio-Visual (AV) systems - which use both sound and video - are very useful for this group.

The so-called "cocktail party problem" [1] arises when multiple talkers are active simultaneously, making it exceedingly difficult for both human listeners and automated systems to separate a target speaker. While purely audio-based methods have matured significantly over the past decade, they remain brittle in realistic conditions. Incorporating visual cues, particularly lip movements, has emerged as a best strategy to overcome the separation problem.

This survey reviews the state-of-the-art in deep-learning-based AV speech enhancement, identifies key research gaps, and outlines a proposed system

tailored specifically for deaf and hard-of-hearing users operating in multi-speaker, real-world environments.

II. BASE PAPER AND EARLY AV MODELS

A. Looking to Listen at the Cocktail Party (Ephrat et al., 2018)

Ephrat et al. [1] presented the foundational model that motivated this work. Their neural network accepts a mixed audio stream alongside the face video of a target speaker - focusing on lip region-of-interest (ROI) features - and separates the target voice from background interference. Trained on AVSpeech, a large-scale dataset comprising thousands of hours of video, the model demonstrated substantially superior performance relative to audio-only baselines. This paper showed that lip movements can help identify and separate one speaker from a noisy mix, even when the system has never seen that speaker before.

B. Audio-Visual Deep CNN (Hou et al., 2018)

Hou et al. [2] created a model called AVDCNN. It uses two separate CNN branches - one for audio and one for video. These branches process the information in parallel, and their features are combined before the final step.

The model also tries to reconstruct the original video while separating the clean speech. This extra task gave small but steady improvements compared to audio-only models. It shows that using both audio and visual information helps, even in simple models.

III. LIP-READING AND SURVEY APPROACHES

A. Lip-Reading Driven Speech Enhancement (Adeel et al., 2019)

Adeel et al. [3] built their system specifically for people with hearing difficulties. It uses a lip-reading front-end to estimate phoneme probabilities. These probabilities then drive a Wiener filter (signal processing technique that removes noise) to enhance the speech.

The system was tested in real noisy environments like cafés and streets. It showed clear improvements in both PESQ (Perceptual Evaluation of Speech Quality) and STOI (Short-Time Objective Intelligibility) scores. This work is highly relevant to our goals.

B. Survey: Michelsanti et al. (2021)

Michelsanti et al. [4] wrote a detailed review of deep learning methods for audio-visual speech enhancement and separation. They grouped the methods based on three main things: how the audio and video are combined (early, late, or hybrid fusion), the type of features used, and the target application.

The survey highlights two important points:

- 1) The lip movements and the sound must be well synchronized for visual information to help effectively.
- 2) Performance drops significantly in real-world multi-speaker situations, especially when the speaker's lips are blocked or not clearly visible.

C. Deep Audio-Visual Speech Recognition (Afouras et al., 2022)

Afouras et al. [5] trained Transformer-based models for audio-visual automatic speech recognition (AV-ASR). Although their main focus was speech recognition rather than speech enhancement, the work showed two important results: visual-only models can already achieve surprisingly good word error rates, and combining audio and visual information gives very robust recognition even in extremely noisy conditions.

The lip-reading features from this research are directly useful for enhancement systems.

IV. TARGET SPEAKER EXTRACTION AND MULTI-MODAL METHODS

A. MuSE (Pan et al., 2020)

Pan et al. [6] introduced MuSE, a multi-modal target speaker extraction system that conditions on lip-video only - eliminating the requirement for a clean reference utterance. This design decision is particularly advantageous for deaf users who cannot readily supply an enrollment audio sample. MuSE generalizes across multiple standard benchmarks and surpasses previous baselines on conventional signal-level metrics.

B. VoiceFormer (Rahimi et al., 2022)

Rahimi et al. [7] extended the classic cocktail-party problem by using three types of input: audio, lip video, and transcribed text.

Their Transformer-based model, called VoiceFormer, can gracefully handle situations where one or more of these inputs are missing or out of sync. This robustness makes VoiceFormer a strong choice for real-world, unconstrained environments.

V. ROBUSTNESS AND REAL-TIME METHODS

A. RAVSS (Pan et al., 2024)

Pan et al. [8] tackled two important limitations of earlier work: handling 2 to 5 speakers talking at the same time, and working well even when the speaker's lips are partially or fully blocked.

Their model, called RAVSS, uses a special occlusion detector. This detector automatically switches between visual-enhanced mode (when lips are visible) and audio-only mode (when lips are not visible).

Tests on the VoxCeleb2 and LRS3 datasets showed consistent improvements over previous single-speaker and occlusion-ignoring systems.

B. RAVEN (Ma et al., 2025)

Ma et al. [9] proposed RAVEN, a model specifically designed for real-time use.

They took pre-trained visual features (learned from lip-reading and speaker recognition tasks) and fine-tuned them inside a causal, streaming separation network. This allows the system to process audio and video continuously without looking into the future.

RAVEN runs smoothly on a standard CPU with less than 50 ms latency per frame. This makes it practical for integration into wearable assistive devices for daily use by deaf and hard-of-hearing people.

VI. COMPARISON OF METHODS

Table I summarizes the key characteristics of the surveyed methods along dimensions relevant to assistive technology deployment.

Method	Year	Multi Speaker	Occlusion Robust	Real Time	Deaf Focused
AVDCNN [2]	2018	No	No	No	No
Ephrat et al. [1]	2018	Yes	No	No	No
Adeel et al. [3]	2019	No	No	No	Yes
MuSE [6]	2020	Yes	Partial	No	No
VoiceFormer [7]	2022	No	Partial	No	No
RAVSS [8]	2024	Yes (2-5)	Yes	No	No
RAVEN [9]	2025	Partial	No	Yes	No
Proposed	-	Yes (2-5)	Yes	Yes	Yes

TABLE I. COMPARISON OF SURVEYED AV SPEECH ENHANCEMENT METHODS

VII. RESEARCH GAPS

Analysis of the surveyed literature reveals five persistent challenges that remain inadequately addressed:

- 1)Limited multi-speaker capacity. The majority of prior systems process a single target speaker at a time. Simultaneous multi-speaker separation remains largely unsolved, with only RAVSS providing a principled solution for up to five concurrent talkers.
- 2)Visual occlusion fragility. Real-world scenarios routinely involve head turns, low illumination, and face coverings. Most models assume frontal, well-lit lip visibility, making them brittle in practice.
- 3)Insufficient focus on assistive technology. Although these systems are clearly useful for deaf and hard-of-hearing people, most of them are designed for general noise reduction. They do not optimize for the perceptual intelligibility metrics that actually matter to hearing-impaired users.
- 4)Computational infeasibility on edge devices. Most high-performing models need powerful server-grade GPUs to run. This makes them unsuitable for low-power devices (smartphones) that deaf and hard-of-hearing users actually need for everyday use.
- 5)Dataset bias toward Western English. Nearly all benchmarks use native English from Western environments. Indian-accented English and local urban noise profiles remain underrepresented.

VIII. PROPOSED SYSTEM

Our proposed system combines useful ideas from previous research to address the five key gaps mentioned earlier. Specifically:

(i) We use visual conditioning from MuSE [6], which allows enrollment using only lip video - no clean reference audio is needed.

(ii) For multi-speaker separation, we adopt the architecture from RAVSS [8], which can separate up to five speakers talking at the same time.

(iii) For real-time performance, we follow the pre-trained visual features and causal streaming design from RAVEN [9], enabling low-latency operation on a standard CPU.

(iv) Occlusion handling is based on the visibility-aware switching mechanism from RAVSS [8], so the system continues working even when lips are partially or fully blocked.

(v) We will evaluate the system on Indian-accented English speakers in real noisy settings (cafés, public transport, markets), following the practical approach of Adeel et al. [3].

IX. CONCLUSION

This survey has followed the development of audio-visual speech enhancement, starting from the work of Ephrat et al. [1] in 2018 up to the latest methods like RAVSS [8] and RAVEN [9]. Overall, the field has moved toward systems that use multiple modalities, handle multiple speakers, and run in real time.

However, important gaps still remain - especially for handling lip/audio occlusion, and working well in non-Western noisy environments.

Our proposed system aims to fill these gaps. It brings together the best parts of the reviewed methods into one complete, efficient pipeline designed for deaf and hard-of-hearing users in multi-speaker settings.

Future work will include collecting new datasets, integrating the models, and running studies to test each design choice.

X. REFERENCES

- [1] A. Ephrat, I. Mosseri, O. Lang, T. Dekel, K. Wilson, A. Hassidim, W. T. Freeman, and M. Rubinstein, "Looking to listen at the cocktail party: A speaker-independent audio-visual model for speech separation," *ACM Trans. Graph.*, vol. 37, no. 4, pp. 1–11, Aug. 2018.
- [2] J.-C. Hou, S.-S. Wang, Y.-H. Lai, J.-C. Lin, Y. Tsao, H.-W. Chang, and H.-M. Wang, "Audio-visual speech enhancement using multimodal deep convolutional neural networks," *IEEE Trans. Emerg. Top. Comput. Intell.*, vol. 2, no. 2, pp. 117–128, Apr. 2018.
- [3] A. Adeel, M. Gogate, A. Hussain, and C. Whittle, "Lip-reading driven deep learning approach for speech enhancement," *IEEE Trans. Emerg. Top. Comput. Intell.*, vol. 5, no. 3, pp. 481–490, Jun. 2021.
- [4] D. Michelsanti, Z.-H. Tan, S.-X. Zhang, Y. Xu, M. Yu, D. Yu, and J. Jensen, "An overview of deep-learning-based audio-visual speech enhancement and separation," *IEEE/ACM Trans. Audio, Speech, Lang. Process.*, vol. 29, pp. 1368–1396, 2021.
- [5] T. Afouras, J. S. Chung, A. Senior, O. Zisserman, and A. Zisserman, "Deep audio-visual speech recognition," *IEEE*

Trans. Pattern Anal. Mach. Intell., vol. 44, no. 12, pp. 8717–8727, Dec. 2022.

- [6] Z. Pan, M. Tao, C. Xu, and H. Li, “MuSE: Multi-modal target speaker extraction with visual cues,” arXiv:2010.07775, Oct. 2020.
- [7] A. Rahimi, S. Liao, H. Hermansky, and B. Raj, “Reading to listen at the cocktail party: Multi-modal speech separation,” in Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR), Jun. 2022, pp. 21–30.
- [8] T. Pan, W. Tao, K. He, and Y. Ma, “RAVSS: Robust audio-visual speech separation in multi-speaker scenarios with missing visual cues,” arXiv:2407.19224, Jul. 2024.
- [9] T. A. Ma, S. Chen, and Y. Luo, “Real-time audio-visual speech enhancement using pre-trained visual representations,” arXiv:2507.21448, Jul. 2025.